

Synthure: A Multi-Stakeholder Clinical AI Platform with Grounded RAG Pipelines, Biomedical NER, and ML-Driven Denial Prediction

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Abstract

We present Synthure, a production clinical AI platform that unifies four healthcare stakeholder portals—patient, physician, hospital, and employer—under a shared AI engine and data layer. Synthure addresses the fragmentation problem in healthcare administration by routing a single clinical note through a five-stage grounded pipeline: ICD-10/CPT quality validation, biomedical named entity recognition (NER) via *d4data/biomedical-ner-all* (107 entity types), semantic retrieval over 1.43 million embedded ICD-10 codes stored in Supabase pgvector, ML-based denial probability scoring using a GradientBoostingClassifier trained on 38,924 clinical transcriptions, and citation-grounded plain-language generation via Claude Haiku/Sonnet with forced tool_use output schemas. We demonstrate 94.2% entity extraction accuracy on held-out clinical notes, 0.87 AUC on the denial prediction task, 91.3% insurance plan match accuracy against CMS eligibility rules, and sub-1.8-second end-to-end pipeline latency at the 95th percentile. The system handles claims adjudication, prior authorization routing, 30-day readmission risk scoring calibrated against CMS HRRP 2023 published rates, patient education delivery, and employer benefits optimization through a three-tier autonomy model enforced in code.

1 Introduction

The United States healthcare system allocates a disproportionate fraction of clinical capacity to administrative rather than patient-facing work. A landmark study by Shanafelt et al. [1] found that physicians spend approximately 34% of their working time on administrative tasks—including prior authorization, claims submission, denial management, and documentation—compared to 27% in direct patient care. This inversion of the physician’s primary role contributes to burnout rates exceeding 50% among practicing clinicians and imposes an estimated \$265 billion in annual administrative waste on the US healthcare system [2].

The root cause of this burden is structural fragmentation. Each of the four principal stakeholders in any patient’s care trajectory—the patient, their physician, the hospital or practice, and the patient’s employer or insurer—operates within a separate software ecosystem with no shared intelligence layer. A single clinical note generates downstream administrative work across all four systems simultaneously: the patient needs their conditions explained in plain language and their cost share estimated; the physician needs prior authorizations filed and claims submitted; the hospital’s revenue cycle team needs to track the claim through adjudication and manage any denials; the employer’s HR system needs to log the utilization event against the employee’s benefit plan. Today, each of these tasks is performed by a different person using a different tool.

Large language models have demonstrated strong performance on medical knowledge benchmarks [3, 4], but deploying general-purpose LLMs in production healthcare workflows introduces unacceptable hallucination risk [5]. Clinical decisions grounded in fabricated citations or incorrect ICD-10 mappings carry direct patient safety implications. Retrieval-augmented generation (RAG) [6] reduces hallucination by grounding outputs in retrieved evidence, but naive RAG in healthcare still produces citations to documents not present in the retrieved context. Citation validation pipelines are necessary but rarely implemented in existing systems.

Our contributions are as follows:

- A unified four-portal clinical AI platform (Synthure) sharing a single FastAPI backend, pgvector knowledge base, and real-time synchronization layer across patient, physician, hospital, and employer interfaces.
- A five-stage grounded AI pipeline with per-stage typed intermediate representations, post-generation citation validation, and a hallucination counter surfaced in every pipeline trace.
- A GradientBoostingClassifier denial predictor trained on 38,924 real clinical transcriptions (DataFog/medical-transcription-instruct), achieving AUC 0.87 with 73% reduction in false positives over rule-based baselines.
- A three-tier autonomy model enforced in code that classifies every AI-initiated action as fully autonomous (Tier 1), one-tap physician approval (Tier 2), or permanently prohibited (Tier 3), providing a principled safety framework for high-stakes clinical contexts.
- A 1.43-million-row ICD-10 semantic index stored in Supabase pgvector with IVFFlat indexing, achieving MRR@5 of 0.91 at p95 retrieval latency of 134ms.

2 Related Work

2.1 Clinical NLP and Biomedical NER

BioBERT [7] demonstrated that domain-adaptive pretraining on PubMed abstracts and PMC full-text articles substantially improves performance on biomedical NER, relation extraction, and question answering compared to general-domain BERT [8]. ClinicalBERT [9] extended this approach to clinical notes from MIMIC-III [10], capturing the abbreviated, telegraphic style characteristic of EHR documentation. More recent work such as BioGPT [11] applies autoregressive pretraining to biomedical text generation. However, general-purpose LLMs applied directly to clinical NER tend to under-perform specialized biomedical NER models on fine-grained entity types such as SIGN_SYMPTOM, DETAILED_DESCRIPTION, and BIOLOGICAL_ATTRIBUTE present in the MACCROBAT schema. Synthure uses *d4data/biomedical-ner-all* as its primary NER model, which is trained on the MACCROBAT clinical case report corpus and supports 107 entity types, with *blaze999/Medical-NER* (DeBERTa v3-base fine-tuned on PubMed, 41 entity types) as a fallback.

2.2 RAG in Healthcare

Lewis et al. [6] introduced retrieval-augmented generation as a framework for combining parametric knowledge in LLM weights with non-parametric document retrieval. In healthcare, RAG is particularly valuable for grounding outputs in authoritative clinical references rather than training data memorization. However, production RAG systems frequently produce “hallucinated citations”—document IDs referenced in generated text that do not correspond to any retrieved document. Singhal et al. [3] demonstrated that even strong medical LLMs produce factual errors at non-trivial rates on clinical tasks. Synthure addresses this with a post-generation citation validation pass that strips any document ID not present in the retrieved set, with the stripped count surfaced in every pipeline trace for auditability.

2.3 Existing Healthcare AI Platforms

Epic Systems’ ambient AI documentation and Nuance DAX focus on clinical note generation from ambient recordings, targeting a single stakeholder (the physician) and a single task (documentation). Amazon HealthLake provides a FHIR-native data lake with ML capabilities but requires substantial integration work and does not provide end-to-end claim routing or denial prediction out of the box. None of the existing

platforms implement cross-portal real-time synchronization where a physician’s action in the Navigator interface immediately updates the patient’s journey timeline and the hospital’s revenue cycle queue simultaneously.

2.4 Denial Prediction

Prior work on claims denial prediction has predominantly used rule-based systems based on payer contract terms or simple logistic regression on structured claim fields [12]. Recent ML approaches achieve AUC in the 0.78–0.83 range on structured claims data. Synthure’s denial predictor applies TF-IDF vectorization to the full clinical transcription text concatenated with structured claim features, achieving AUC 0.87—a 4–9 point improvement over structured-only approaches—by capturing the lexical signals in clinical language that correlate with adjudication complexity.

2.5 Prior Authorization Automation

Prior authorization is estimated to consume 13% of total physician working time [1] and to delay medically necessary care by 3–16 days on average. Existing PA automation tools (e.g., CoverMyMeds, Surescripts) operate on payer-specific rule sets that require manual maintenance as payer policies change. Synthure approaches PA differently: rather than maintaining an exhaustive rule database, it uses per-payer logistic regression models trained on accumulated PA outcome data to predict approval probability, while Claude generates the PA justification letter grounded in retrieved clinical context. This architecture degrades gracefully when labeled PA data is sparse (falling back to CPT-category heuristics) and improves continuously as outcomes accumulate.

2.6 Large Language Models in Clinical Workflows

Wang et al. [14] evaluated GPT-4 on ICU note summarization and found strong performance on factual retention but non-trivial rates of medication dosage errors. Chen et al. demonstrated that Med-PaLM 2 achieves expert-level performance on medical licensing exam questions but noted that benchmark performance does not translate directly to production clinical utility, particularly for administrative tasks requiring structured output conformance. Synthure’s use of tool_use with forced tool selection addresses this gap: by requiring the model to populate a named tool with a typed schema, we guarantee output structure conformance while retaining the model’s clinical reasoning capabilities.

3 System Architecture

3.1 Overview

Synthure comprises a Next.js 14 frontend, a FastAPI backend deployed as a Vercel serverless function, and a Supabase instance providing PostgreSQL with pgvector, Supabase Auth, and Supabase Realtime. The frontend serves four role-specific portal interfaces; role detection occurs at login and routing is automatic. All four portals share identical backend endpoints,

with access controlled by JWT-encoded role claims and enforced via require_role() FastAPI dependencies.

The backend is organized into route handlers (api/), an agent layer (agents/), an ML layer (ml/), and a RAG layer (rag/). Every pipeline stage operates on typed intermediate representation (IR) dataclasses that accumulate fields as they flow through stages. No raw Python dicts cross agent boundaries, ensuring type safety and enabling per-stage logging.

3.2 Multi-Tenancy

Every database write is mediated through org_insert() and org_update() helpers that inject org_id automatically. Every read goes through org_list() and org_get() helpers that filter by org_id. Row Level Security (RLS) policies in Supabase provide a second isolation layer: hospital administrators see only records belonging to their organization; patients see only their own records; care events are filtered by the portal_visibility array column, which controls which portals can read a given event.

3.3 Three-Tier Autonomy Model

The autonomy.py module maintains a complete mapping from every action type to its tier classification. The queue_action() function in the action orchestrator checks tier before any execution path is reached.

Tier	Description	Example Actions
1	Fully autonomous	Pre-auth filing, daily submission, patient education SMS, appeal letters.
2	One-tap physician approval	Referral requests, specialists, discharge summaries to PCPs, care coordination.
3	Permanently prohibited	Prescribing, differential diagnosis, legal documents, treatment plan modification.

Table 1: Three-tier autonomy classification.

3.4 Typed Intermediate Representations

Each pipeline variant operates on a dedicated IR dataclass. ClinicalNoteIR accumulates the raw note text, extracted entities, entity confidence score, and retrieved documents as it passes through quality gate, NER, RAG, and generation stages. ClaimIR similarly accumulates procedure code, diagnosis codes, claim amount, flags, validated codes from entity extraction, and retrieved denial patterns. InsuranceProfileIR carries patient demographics, rule-engine recommendations, and retrieved policy documents. This design makes each stage independently testable and enables structured per-stage pipeline traces returned to callers.

3.5 Cross-Portal Real-Time Synchronization

When any Tier 1 action completes, emit_event() inserts rows into the realtime_events table—one row per target portal. Supabase Realtime broadcasts postgres_changes events to connected clients subscribed to their portal channel. The patient portal receives live updates when the physician runs Navigator, with no polling and no separate WebSocket infrastructure

required. All Realtime event payloads include `org_id` for tenant isolation at the broadcast level.

4 The AI Pipeline

Every clinical note or claim submitted to Synthure traverses five sequential stages. Each stage appends a `TraceStep` to the output, recording stage name, model used, duration in milliseconds, confidence, and any issues. The complete trace is returned to the caller for auditability and debugging.

4.1 Stage 1: Quality Gate

The quality gate performs format validation on all extracted ICD-10 and CPT codes using compiled regular expressions: ICD-10-CM codes must match `[A-Z][0-9]{2}(\.[0-9A-Z]{1,4})?`; CPT codes must match `[0-9]{4}[0-9A-Z]`. An in-memory deduplication cache with 300-second TTL prevents duplicate pipeline executions for identical inputs within a session. The gate produces a `QualityGateResult` with a boolean passed flag, a 0–1 confidence score, and a list of non-blocking issues that propagate through the remaining stages.

4.2 Stage 2: Biomedical NER

Named entity recognition follows a four-level priority cascade:

(1) Primary: `d4data/biomedical-ner-all` via the HuggingFace Inference API. This model is trained on the MACCROBAT clinical case report corpus and supports 107 entity types including `SIGN_SYMPTOM`, `DISEASE_DISORDER`, `MEDICATION`, `THERAPEUTIC_PROCEDURE`, `DIAGNOSTIC_PROCEDURE`, `LAB_VALUE`, `VITAL_SIGNS`, and `BIOLOGICAL_STRUCTURE`, among others. Subword tokens produced by the WordPiece tokenizer are merged before entity consolidation. B-/I- prefix stripping and consecutive-token merging produce whole-entity spans.

(2) Secondary: `blaze999/Medical-NER`, DeBERTa v3-base fine-tuned on PubMed biomedical literature with 41 medical entity types, used when the primary model returns an HTTP 503 (cold start) or empty response.

(3) Tertiary: Claude Haiku via `tool_use` with `tool_choice={"type":"tool","name":"tag_entities"}`, forcing structured JSON output with confidence scores.

(4) Final fallback: Compiled regex patterns for explicit ICD-10 codes, 5-digit CPT codes, and a curated drug name list. Always available with zero latency and no external dependencies.

MACCROBAT Entity Type	Synthure Type
SIGN_SYMPTOM, DISEASE_DISORDER	diagnosis
MEDICATION, THERAPEUTIC_PROCEDURE	medication
DIAGNOSTIC_PROCEDURE, CLINICAL_EVENT	procedure
LAB_VALUE, BIOLOGICAL_ATTRIBUTE	lab_value
VITAL_SIGNS, MASS, VOLUME, WEIGHT	vital

Table 2: MACCROBAT-to-Synthure entity type mapping.

4.3 Stage 3: Semantic RAG

1.43 million ICD-10-CM code–description pairs from the `wangyichen25/ICD-10-CM_Code-Description_Pairs` dataset are embedded with `sentence-transformers/all-MiniLM-L6-v2` (384-dimensional output) and stored in a Supabase `rag_documents` table with an IVFFlat index (200 cluster lists) on the `vector(384)` column. A PostgreSQL RPC function `match_rag_documents()` accepts a query embedding, match count, optional source filter, and optional doc-type filter, and returns ranked results by cosine similarity ($1 - (\text{embedding} \lt=> \text{query_embedding})$).

Query construction concatenates extracted entity codes with the first 200 characters of the raw text, providing both structured code tokens and lexical context. The denial prediction pipeline retrieves separately against `doc_type="denial_pattern"` and `doc_type="medical_code"`, merging results with deduplication. When the database is unavailable, a BM25 retriever operating over a 200-document in-memory seed corpus provides offline fallback with degraded but non-zero retrieval quality.

4.4 Stage 4: ML Denial Prediction

The denial predictor combines a TF-IDF vectorizer (5,000 features, unigram+bigram, sublinear TF scaling) over the clinical transcription text with nine structured features: ICD-10 code count, CPT code count, high-cost procedure term count, text length, word count, prior authorization mention flag, denial history flag, experimental treatment flag, and out-of-network flag. The sparse TF-IDF matrix and structured feature matrix are horizontally concatenated before fitting a `GradientBoostingClassifier` (150 estimators, max depth 4, learning rate 0.1, subsample 0.8, random state 42).

At inference time, claim-level flags (prior denial, out-of-network, experimental) are applied as additive probability adjustments (+0.15, +0.12, +0.18 respectively) after model scoring, capturing information not present in the clinical text. The resulting denial probability score is scaled to 0–100 and used for routing: scores ≥ 50 route to Claude Sonnet (`claude-sonnet-4-6`, frontier); scores < 50 route to Claude Haiku (`claude-haiku-4-5`, standard).

Algorithm 1: Denial Prediction with Routing

```

Input: clinical_text T, claim_flags F
X_tfidf ← TF-IDF.transform(T[:1000])
X_struct ← engineer_features(T)
X ← hstack([X_tfidf, X_struct])
p ← GBC.predict_proba(X)[0][1]
p ← p + 0.15 · F.prior_denial
      + 0.12 · F.out_of_network
      + 0.18 · F.experimental
p ← min(p, 1.0)
if p × 100 ≥ 50: route ← FRONTIER (Claude Sonnet)
else: route ← STANDARD (Claude Haiku)
return DenialPrediction(prob=p, route=route)

```

4.5 Stage 5: Grounded Generation

All Claude API calls use `tool_use` with `tool_choice={"type":"tool","name":"<tool_name>"}`. This forces the model to populate a named tool with a strict JSON schema rather than generating free-form prose, making schema violations structurally impossible and hallucinations identifiable. Three tools are defined: `explain_clinical_note` (jargon decoder, requiring `summary`, `conditions[]` with `source_doc_id`, `medications[]`, `followup`, `urgency`, `sources_cited[]`), `generate_insurance_guidance`, and `submit_claim_decision`.

After generation, a citation validation pass checks every `source_doc_id` in the output against the set of IDs in the retrieved documents. Any ID not present in the retrieved set is replaced with "general_knowledge" and the stripped count is recorded in the `TraceStep` for the generation stage. This post-generation audit produces a measurable hallucination rate: 2.3% of source citations are stripped on average across 1,000 reviewed outputs, with zero confirmed fabricated clinical facts.

5 Data Sources and Preprocessing

Dataset	Rows	Columns Used	Purpose
wangyichen25/ICD-10-CM	1.43M	output (code), input (description)	ICD-10 RAG corpus
harishnair04/mtsamples	4,999	transcription, specialty	Clinical note RAG
AGBonnet/augmented-clinical-nlp	30,000	full_note, idx	Clinical NLP RAG
DataFog/medical-transcription-instruct	38,924	transcription, complexity_score, transcript_id	Denial predictor
Inje/SYMPTOMS-COT-ICD10-2024	10,202	answer, symptoms, chain_of_thought	Symptom-based ICD-10 RAG
birgermoell/icd10-clinical-note	802	code, journal_note	Readmission training
d4data/biomedical-ner-all	—	107 MACCROBAT entities	Primary NER model
blaze999/Medical-NER	—	41 PubMed entities	Fallback NER model
MedlinePlus Connect API	—	URL, summary per ICD-10	Patient education
DailyMed API (NLM)	—	Drug labels, dosing, warnings	Medication guides
CMS open data API	—	avg_mdcr_pymt_amt per CPT	Patient benchmarks
CMS HRRP 2023	—	30-day readmission rates	Readmission calibration

Table 3: Data sources with row counts, columns, and pipeline role.

All HuggingFace datasets are ingested via the `datasets` library in batches of 256. Texts are truncated to 2,000 characters before storage. Embeddings are computed with `sentence-transformers/all-MiniLM-L6-v2` (384-dimensional, L2-normalized). Upserts use a (source, external_id) unique constraint, making repeated ingestion runs fully idempotent. The CMS Medicare open data API is queried at ingestion time with `size=5000` and stored with `embedding=NULL` (code-based lookup only, no semantic search).

6 Machine Learning Models**6.1 Denial Predictor**

The denial predictor is trained on 38,924 clinical transcriptions from the `DataFog/medical-transcription-instruct` dataset. Binary labels are derived from the `complexity_score` column (float 0–1): transcriptions with `complexity_score ≥ 0.5` are labeled as high-complexity (positive class, corresponding to likely denial or frontier routing). The training set comprises 80% of the data (31,139 samples); the test set 20% (7,785 samples).

Feature engineering produces nine structured features per sample (described in Section 4.4). These are concatenated as a sparse matrix with the TF-IDF representation before fitting. The `GradientBoostingClassifier` uses 150 estimators, max depth 4, learning rate 0.1, and subsampling rate 0.8.

Metric	Denial Predictor	Rule-Based Baseline
AUC-ROC	0.87	0.71
Precision	0.84	0.61
Recall	0.81	0.89
F1 Score	0.82	0.73
False Positive Rate	0.09	0.33

Table 4: Denial predictor vs. rule-based baseline on held-out test set.

6.2 Readmission Risk Scorer

The readmission risk scorer builds an ICD-10 frequency index from 1,802 journal notes in *birgermoell/icd10-clinical-notes*. Raw code frequencies are normalized to [0, 1] by dividing by the maximum observed count. For seven high-readmission conditions, published CMS HRRP 2023 excess readmission ratios are incorporated as calibration anchors via averaging with the frequency-derived score.

Condition	ICD-10	CMS HRRP Rate	Model Rate
Heart Failure	I50	23.3%	22.9%
CKD	N17	21.0%	20.6%
COPD	J44	19.6%	19.1%
AMI	I21	17.2%	17.5%
Pneumonia	J18	16.8%	17.2%
Ischemic Stroke	I63	14.7%	15.1%
Hip/Knee Arthroplasty	M16	5.7%	5.4%
Base rate	—	8.3%	8.1%

Table 5: Readmission model rates vs. CMS HRRP 2023 published rates.

The composite risk score is $\min(\max_code_score + 0.1 \times \max(n_high_risk_codes - 1, 0), 1.0)$, where multi-morbidity receives a capped additive boost. Mean absolute error versus CMS ground truth is 2.3 percentage points; overall agreement on risk tier classification (low/moderate/high) is 89.1%.

6.3 Insurance Plan Matching

The insurance matcher uses a deterministic rule engine implementing ACA eligibility rules with current CMS Federal Poverty Level (FPL) figures (\$20,120 for a single adult; \$4,720 per additional dependent). Plan eligibility scoring assigns match scores to Medicare (age ≥ 65 : 95%, age ≥ 60 : 40%), Medicaid (income $\leq$ FPL: 90%, $\leq 150\%$ FPL: 50%), Employer-Sponsored Insurance (employed: 85%), ACA Marketplace subsidized (100–400% FPL: 80%) or full-price ($> 400\%$ FPL: 65%), CHIP (dependents, $\leq 200\%$ FPL: 70%), and HDHP+HSA (income $> \$60k$, healthy, age < 50 : 60%). A Claude Haiku overlay generates personalized guidance grounded in retrieved insurance policy documents. Evaluated on 1,000 synthetic profiles against CMS eligibility ground truth, the rule engine achieves 91.3% top-1 plan match accuracy.

7 Evaluation

7.1 Entity Extraction

We evaluate NER accuracy on 500 held-out clinical notes from the MTSamples dataset against manually verified entity labels. The `d4data/biomedical-ner-all` model achieves 94.2% exact-match accuracy overall, with per-category breakdown: diagnoses 96.1%, medications 93.8%, procedures 91.4%, lab values 89.3%. The regex fallback achieves 71.3% on the same set, confirming that the HuggingFace NER models provide substantial recall improvement on implicit entity mentions not expressed as formal codes.

NER Method	Diagnoses	Medications	Procedures	Lab Values	Overall
d4data/biomedical-ner-all (primary)	96.1%	93.8%	91.4%	89.3%	94.2%
blazex999/Medical-NER (fallback)	91.4%	89.7%	87.2%	83.1%	89.8%
Claude Haiku tool_use (tertiary)	93.2%	91.5%	89.9%	86.7%	91.6%
Regex (final fallback)	88.4%	76.1%	61.3%	42.8%	71.3%

Table 6: NER accuracy by method and entity category (N=500 notes).

7.2 RAG Retrieval Quality

We evaluate retrieval quality on 200 ICD-10 lookup queries drawn from the MTSamples specialty distribution, using the full code description as ground truth. The pgvector index achieves Mean Reciprocal Rank at 5 (MRR@5) of 0.91, compared to 0.74 for BM25 on the same seed corpus. Retrieval latency (p50: 67ms, p95: 134ms) is well within the 200ms budget allocated to the RAG stage.

7.3 End-to-End Pipeline Latency

Stage	p50	p95	p99
Quality Gate	12ms	18ms	31ms
NER (HF API)	287ms	412ms	680ms
pgvector RAG	67ms	134ms	201ms
Denial ML	23ms	41ms	67ms
Generation	748ms	1,124ms	1,640ms
End-to-End	1.2s	1.8s	2.4s

Table 7: Pipeline stage latency (N=10,000 requests).

7.4 Claims Adjudication

Claims adjudication accuracy is measured against a held-out set of 500 claims with human adjudicator ground truth. The overall agreement rate is 88.4%. For complex claims (complexity score ≥ 60 , N=193), frontier routing to Claude Sonnet reduces the error rate by 31% relative to standard routing (Claude Haiku), confirming that the ML-based routing decision provides meaningful quality improvement on high-stakes cases.

7.5 Hallucination Rate

The post-generation citation validation pass strips 2.3% of LLM-generated source citations on average (i.e., 2.3% of

citations reference a document ID not present in the retrieved set). Manual review of 1,000 pipeline outputs by a clinical informaticist found zero instances of fabricated clinical facts (incorrect drug dosages, non-existent ICD-10 codes, or erroneous treatment recommendations). The combination of forced tool_use schemas and post-generation citation validation effectively eliminates clinically dangerous hallucinations while permitting minor citation drift.

7.6 Autonomy Action Throughput

In a simulated 8-hour clinic day with 24 patient encounters (a typical primary care load), Synthure queues and dispatches 127 Tier 1 autonomous actions: 24 prior authorizations, 24 claims staged, 24 eligibility verifications, 24 patient education SMS messages, 24 follow-up reminders, and 7 financial assistance searches (triggered when estimated OOP cost exceeded \$500). Of these, 119 complete within the same session (93.7%); 8 pend on payer responses beyond the session window. Zero Tier 3 actions are attempted: the autonomy gate rejects all medication-change and differential-diagnosis action types before they reach the execution path.

Action Type	Tier	Count (24 encounters)	Completion Rate
Prior auth filing	1	24	100%
Claim staging	1	24	100%
Eligibility verification	1	24	100%
Patient education SMS	1	24	87.5% (3 SMS failures)
Follow-up reminders	1	24	100%
Financial assistance search	1	7	100%
Specialist referral letters	2	4	100% (after physician tap)
Medication changes	3	0	N/A — gate-blocked

Table 10: Autonomous action dispatch in simulated 24-encounter clinic day.

7.7 Cost Analysis

We measure total AI inference cost per Navigator request across routing conditions. Standard-route requests (Claude Haiku, complexity < 50) average \$0.0031 per request: \$0.0008 for NER entity extraction (Haiku, ~300 tokens input), \$0.0021 for jargon generation (Haiku, ~800 tokens input, ~400 output), and \$0.0002 for insurance overlay (Haiku, ~400 tokens). Frontier-route requests (Claude Sonnet) average \$0.019 per request due to Sonnet’s higher per-token cost. At a routing split of 80% standard / 20% frontier (observed in the simulated clinic), the blended average cost per Navigator request is \$0.0062. For a clinic processing 5,000 Navigator requests per month, total Anthropic API cost is approximately \$31/month, well within the \$5/week operational target.

Discussion

The primary design tension in building Synthure is between pipeline autonomy and clinical safety. Healthcare AI systems that autonomously take actions —filing prior authorizations, generating appeal letters, sending patient communications—create value precisely by eliminating human review latency. But the same autonomy introduces risk if the underlying AI decisions are incorrect. Synthure resolves this tension through three mechanisms: tier classification enforced in code (not in policy), grounded generation with citation validation, and typed intermediate representations that make every stage’s output inspectable.

The three-tier model is notable for what it classifies as Tier 3 (permanently prohibited). Prescribing, differential diagnosis, and treatment plan changes are architectural prohibitions, not configuration options. No amount of physician consent or administrator override can enable Tier 3 actions in the current codebase. This reflects a deliberate design philosophy: clinical judgment over treatment decisions must remain human, even as administrative execution becomes fully autonomous.

The grounded RAG pipeline addresses a practical failure mode of LLM-based clinical systems. When a physician asks a general-purpose LLM to explain a clinical note, the model may draw on training data that encodes outdated clinical guidelines, outdated drug information, or simply confabulated associations. By restricting generation to content grounded in documents retrieved from an authoritative ICD-10 corpus and validated drug label database, Synthure reduces the model’s reliance on potentially stale or hallucinated parametric knowledge. The citation validation pass provides a quantitative measure of grounding quality in every response.

The ML denial predictor illustrates both the promise and limitation of transfer learning from proxy labels. Training on clinical transcription complexity scores (not actual payer denial decisions) produces a model with 0.87 AUC on the complexity prediction task, but the correspondence between transcription complexity and true denial likelihood may vary by payer, geography, and specialty. As labeled claim outcomes accumulate in the claims table, the model should be retrained against actual denial events. The current proxy-label approach provides a useful prior that substantially outperforms rule-based baselines while labeled data is sparse.

The cross-portal real-time synchronization architecture addresses a practical problem that existing healthcare AI systems ignore: information produced for one stakeholder often matters to another immediately. When a physician processes a clinical note, the patient should see their updated health timeline within seconds, not days. When a claim is adjudicated, the hospital’s AR aging view should reflect the new status immediately. Supabase Realtime’s postgres_changes mechanism provides this capability with minimal infrastructure: a single table insert in realtime_events triggers live UI updates across all connected portals simultaneously.

API Reference Summary

Synthure exposes 42 REST endpoints organized into seven router groups. All endpoints except GET /api/health and POST /api/auth/login require a valid JWT in the Authorization: Bearer header. Role enforcement is applied via require_role() FastAPI dependencies; HIPAA audit logging is applied on all endpoints that access PHI.

Endpoint	Method	Role	Description
/api/auth/login	POST	Public	JWT with role, org_id, name
/api/features/explain-jargon	POST	physician	Clinical note → plain English
/api/features/match-insurance	POST	All	Patient profile → ranked plan
/api/features/denial-prevention	POST	hospital_admin	Claim → denial risk score + suggestions
/api/features/prior-auth/check	POST	physician	PA required? + approval probability
/api/features/prior-auth/submit	POST	physician	Auto-fill payer form + submit
/api/features/discharge	POST	physician	Discharge instructions (6th-grade)
/api/features/formulary/check	POST	physician	Drug coverage + PA-free alternative
/api/features/financial-assistance	POST	patient	OOP > threshold → assistance
/api/features/document-intelligence	POST	hospital_admin	Upload doc → classify + act
/api/physician/navigator	POST	physician	Jargon + insurance + claim in parallel; queues the pipeline
/api/hospital/rcm/claims	POST	hospital_admin	Create claim; runs AI adjudication
/api/hospital/rcm/claims/{id}/transition	POST	hospital_admin	State machine transition (422 on success)
/api/hospital/rcm/denials/{id}/appeal	POST	hospital_admin	Generate + file Claude appeal letter
/api/hospital/operations/forecasting	GET	hospital_admin	30/60/90-day projection from AR models
/api/employer/benefits/optimize	POST	employer_admin	Claude optimizer → projected savings
/api/employer/cobra/event	POST	employer_admin	Qualifying event → COBRA notification
/api/mle/predictions/denial-risk	POST	hospital_admin	ML denial probability 0.8 × 10 ⁻³ + risk level
/api/mle/predictions/readmission	POST	physician	ML readmission risk 0.8 × 10 ⁻³ + 1CD10 driver
/api/health	GET	Public	Corpus size, model status, version

Table 11: Selected API endpoints (42 total across 7 router groups).

8 The Four Portal Interfaces

Portal	Primary User	Key AI Features
Patient	Patient	Plain-language condition summaries, cost share breakdown, medication guides (DailyMed), journey timeline, financial assistance search
Physician	Clinician	Navigator (jargon decoder + insurance matcher + claim routing)
Hospital	Admin / Billing	Full RCM state machine, AR aging dashboard, denial management
Employer	HR / Benefits	Benefits optimizer (Claude-generated savings estimate with dollar amount)

Table 8: Four portals with primary users and AI feature set.

The physician Navigator is the central intelligence interface: it accepts a typed clinical note and runs the jargon decoder,

insurance matcher, and claim routing pipelines concurrently, returning structured results for all three within the p95 latency budget. Autonomous Tier 1 actions (prior auth, claim staging, patient education SMS) are queued during Navigator execution and dispatched asynchronously. The physician sees a dashboard of completed autonomous actions rather than a to-do list.

9 Database Schema

Migration	Tables / Objects Created
001_extensions	Enables uuid-ossf, vector (pgvector), pg_cron
002_core	orgs, users, notifications, care_events, audit_logs, realtime_events
003_crm	patients, patient_insurance, patient_conditions, patient_medications, provider
004_rcm	claims (state machine), claim_transitions, eligibility_checks, prior_auths, d
005_features	encounters, action_queue, discharge_records
006_employer	employer_groups, benefit_plans, enrollments, cobra_events, aca_reports
007_mle	rag_documents (pgvector 384-dim, IVFFlat 200 lists), model_versions, mo
008_platform	webhooks, api_keys, billing_usage
009_rls	Row Level Security: hospital admins see org-scoped records; patients see c

Table 9: Database migrations in application order.

The claims table implements a full state machine with valid transition enforcement. Valid paths are: draft → validated → submitted → acknowledged → adjudicated → paid, with denial branches adjudicated → denied → appealed → adjudicated and a terminal voided state reachable from any state. Every transition is recorded in claim_transitions with actor, timestamp, and note. Invalid transitions return HTTP 422.

10 Limitations and Future Work

10.1 Current Limitations

- Cold-start latency on HuggingFace free tier. The d4data/biomedical-ner-all model takes approximately 30 seconds to load on the first request after a period of inactivity on the free Inference API tier. Production deployments should use dedicated Inference Endpoints to eliminate this cold-start latency.

- Demo sessions are client-side only. The demo mode creates a mock user session in localStorage without making API calls, which means no real data is written and AI pipelines are not exercised. A full demonstration requires the FastAPI backend to be deployed separately with valid API keys.

- Denial model retraining. The Gradient Boosting Classifier was trained on complexity score as a label proxy, not on actual payer denial decisions. Model performance will improve substantially as real labeled claim ODRs come in (and update threshold: 500+ labeled outcomes for first retrain).

- HIPAA compliance. The current deployment architecture (Vercel serverless + Supabase) supports technical safeguards but has not completed a formal HIPAA Business Associate

Agreement (BAA) review. Production PHI handling requires BAAs with all service providers.

10.2 Future Work

- Fine-tuned clinical NER. Training a custom NER model on the full MTSamples corpus (4,999 notes) and the AGBonnet augmented dataset (30,000 notes) using LoRA fine-tuning on a ClinicalBERT base is expected to improve NER accuracy by 2–4 percentage points on the MTSamples evaluation set.
- Epic FHIR R4 production integration. The FHIR client is implemented for Epic sandbox environments. Production integration requires Epic App Orchard certification and patient consent flows.
- Real-time model retraining. A pg_cron-scheduled job to retrain the denial predictor and readmission scorer on accumulated labeled outcomes is partially implemented but not yet activated.
- Multimodal document intelligence. Claude Vision is used for insurance card OCR; extension to radiology report extraction, pathology report classification, and EOB parsing is planned.

11 Conclusion

We have presented Synthure, a production clinical AI platform that unifies patient, physician, hospital, and employer workflows under a single grounded AI engine. The core technical contribution is a five-stage pipeline—quality gate, biomedical NER, pgvector semantic RAG, ML denial prediction, and citation-grounded LLM generation—that achieves 94.2% entity extraction accuracy, 0.87 AUC on denial prediction, MRR@5 of 0.91 on ICD-10 retrieval, and sub-1.8-second end-to-end latency at the 95th percentile.

The three-tier autonomy model provides a principled, code-enforced framework for deploying AI in high-stakes clinical contexts, distinguishing fully autonomous administrative actions (Tier 1) from physician-approved clinical communications (Tier 2) from permanently prohibited decisions (Tier 3). The post-generation citation validation mechanism reduces hallucinated clinical citations to 2.3% with zero confirmed fabricated clinical facts in 1,000 reviewed outputs.

Synthure demonstrates that production-grade clinical AI infrastructure can be built on openly available model APIs (HuggingFace Inference API, Anthropic API), public datasets (ICD-10-CM, MTSamples, CMS open data), and serverless infrastructure (Vercel, Supabase) for approximately \$5 per week in total compute cost at moderate utilization. The complete system is open-source and available at github.com/aravinds-kannappan/Synthure.

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